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Multi-Contrast Human Brain CEST MRI at 11.7 T: First In Vivo Demonstration

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ABSTRACT

Purpose: To implement and evaluate chemical exchange saturation transfer (CEST) at ultra-high field on a whole-body 11.7 T scanner using parallel transmission (pTx).

Methods: Tailored pTx CEST saturation pulses were optimized to achieve a spatially uniform target B_1^+ while satisfying SAR and power constraints either defined by conservatively defined radio-frequency (RF) power limits or by a virtual observation point (VOP)-based framework. The resulting pulse performance at 11.7 T was assessed through numerical simulations and compared against corresponding designs at 7 T. The proposed saturation pulses were integrated into a two-dimensional multi-slice EPI sequence. Experimental testing of the sequence was performed in vitro using metabolite-doped phantoms and in vivo in four healthy volunteers.

Results: The pTx pulse designs optimized under VOP-based SAR constraints at 11.7 T achieved comparable performance to 7 T designs despite higher intrinsic B_1^+ inhomogeneity. In vitro experiments validated the sequence implementation and confirmed detection of the targeted metabolites. In vivo experiments demonstrated acceptable saturation homogeneity ($\text{nRMSE} \leq 12.1\%$) in a slab of thickness 1 cm, with only minimal additional improvement achieved through B_1^+ post-processing correction. Across the cohort, robust gray matter–white matter contrast was consistently observed for multiple CEST contrasts.

Conclusion: VOP-based SAR-constrained pTx CEST saturation pulse design enabled robust and reproducible contrast at 11.7 T by effectively mitigating B_1^+ inhomogeneities while remaining within safety limits, facilitating translation to future in vivo clinical research applications.

1 | Introduction

Chemical exchange saturation transfer (CEST) [1] is a noninvasive MRI contrast mechanism that enables the indirect detection of low-concentration metabolites and mobile proteins by exploiting proton exchange with bulk water. Selective radio-frequency (RF) saturation of exchangeable solute protons at their resonance

frequencies results in the transfer of saturation to bulk water protons, generating CEST contrast through changes in the water signal only. Prolonged saturation amplifies the resulting attenuation of the water signal, thereby enhancing sensitivity to molecular components that are difficult and time-consuming to image using conventional MR spectroscopic imaging [2] while eliminating the need for exogenous tracers, unlike positron emission

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tomography. CEST MRI has been used to probe a variety of molecules, such as amide protons in brain tumors [3–5] or glutamate in the healthy brain at 7 T [6] and is increasingly explored in clinical research in stroke [7], epilepsy [8, 9], and other neurologic or oncologic diseases [10–13]. The launch in 2024 of the investigational whole-body 11.7 T Iseult MR scanner [14] in France (NeuroSpin, CEA) offers a unique opportunity to further explore the capabilities of CEST MRI at extreme magnetic fields.

At high magnetic field strengths, both the longitudinal relaxation time of water and the spectral separation between resonances increase [15], enhancing CEST sensitivity and potentially enabling the detection of a broader range of CEST effects. In this context, glutamate-weighted CEST at 11.7 T is of interest, as its detectability at field strengths below 7 T remains difficult due to its high exchange rate [6, 16]. Hydroxyl-weighted CEST, with resonances as close as 1 ppm to water, may benefit substantially from the increased spectral separation available at ultra-high field [17], as previously demonstrated at 7 T [18].

While ultra-high magnetic field strength provides clear benefits for CEST MRI, it also introduces significant technical challenges. In particular, spatial B_0 and B_1^+ inhomogeneities intensify, resulting in enhanced susceptibility effects, pronounced spatial variation in RF excitation, and increased RF energy deposition. Addressing these limitations has therefore become a major focus of recent methodological developments in ultra-high field CEST MRI.

The most efficient way to mitigate B_1^+ inhomogeneities is to use parallel transmission (pTx) [19], which employs multiple independently driven transmit channels. Dynamic RF shimming with pTx has been widely adopted in ultra-high field MRI, notably for slice-selective spoke excitations [20] and three-dimensional excitation schemes [21] to reduce flip-angle nonuniformity. More recently, pTx has been extended to CEST MRI to improve saturation homogeneity at ultra-high field. The benefits of pTx for CEST imaging were demonstrated using the MIMOSA [22] approach, which employs two complementary alternating static B_1^+ shim configurations to mitigate B_1^+ inhomogeneities. Further improvements were achieved using PUSH [23] saturation pulses, which optimize saturation homogeneity based on the measured B_1^+ distribution. While subject-specific pTx pulse optimization remains time-consuming, universal pulse (UP) [24] implementations of CEST saturation pulses have also been proposed in two-dimensional acquisitions at 7 T [18], paving the way for simpler, faster, and clinically feasible CEST protocols. The utility of such approaches was further demonstrated for whole-brain 3D CEST imaging at 7 T [25, 26], using conservative vendor-defined specific absorption rate (SAR) limits for the pulse design. Delebarre et al. [18] proposed the optimization of CEST saturation pulses using virtual observation points (VOPs) [27], enabling real-time SAR supervision and more efficient use of pTx capabilities at 7 T, resulting in substantial gains in CEST saturation uniformity of up to 4.7 fold.

The present work reports the first implementation and assessment of CEST MRI on a human 11.7 T scanner, combining custom sequence development, numerical simulations, phantom experiments, and in vivo measurements. Tailored CEST pTx pulses and slice-specific three-spoke excitation pulses were

designed to achieve optimal saturation under VOP-based SAR constraints. The performance of the CEST pulses was evaluated through simulations and benchmarked against numerical results at 7 T. Several CEST contrasts were evaluated in realistic head phantoms doped with metabolites and in healthy volunteers. Finally, a B_1^+ post-processing correction [28] was evaluated on the acquired in vivo data.

2 | Methods

2.1 | Experimental Set-Up

All in vivo experiments were performed on the investigational whole-body 11.7 T Iseult MR scanner using an in-house head coil with 8 transmit (Tx) and 31 receive (Rx) channels [29]. In vitro experiments were conducted on the same system, as well as on an investigational whole-body 7 T MR scanner equipped with a Nova head coil (Nova medical, Wilmington, MA, USA) featuring 8Tx-32Rx channels.

For the in vitro experiments, a human head-shaped phantom was fabricated using 3D printed resin [30]. The phantom was initially filled with a phosphate-buffered saline (PBS) solution containing 10 mM L-glutamic acid, 10 mM creatine, and 20 mM D-glucose. To approximate the electrical and relaxation properties of human brain tissue, 4 g/L sodium chloride and 0.064 mM gadoterate meglumine (Dotarem) were added. To prevent bacterial degradation, 4 mM sodium azide was also included. The pH was adjusted to 7, and the phantom was heated overnight at 37°C prior to the experiments.

Subsequently, the phantom was emptied and refilled with 3% porcine skin gelatin to enable APT-weighted CEST contrast, while the quantities of all other components remained unchanged relative to the PBS-based phantom.

For the in vivo experiments, four healthy volunteers (26 ± 5 years old, two males and two females) were scanned after providing written informed consent, in accordance with a French ethics committee.

2.2 | CEST MRI Methodology

2.2.1 | Tailored pTx Pulse Design and Implementation

Tailored pTx CEST saturation and excitation pulses were designed for each acquisition, for both in vitro and in vivo experiments.

To this end, a B_0 field map was acquired using a three-dimensional gradient-recalled echo (GRE) sequence, followed by B_1^+ field maps acquired with an interferometric turbo-FLASH sequence [31], with 2.5 mm isotropic resolution and $5 \times 5 \times 2.5 \text{ mm}^3$ resolution with a 2.5 mm slice gap, respectively (matrix size $128 \times 128 \times 72$ and $40 \times 64 \times 40$, respectively). B_0 shimming was performed at the beginning of each acquisition session using in-house software based on quadratic programming [32] specifically designed for ultra-high field acquisitions.

The CEST pulse optimization followed the methodology described by Delebarre et al. [18] and consisted in designing pTx Gaussian saturation pulses composed of two complementary modes (A and B) to achieve effective saturation over a 1 cm axial slab positioned at the scanner isocenter. The two modes were applied in an alternating fashion and repeated N times until the desired total saturation time (T_{sat}) was reached. The optimization minimized the L_2 norm between the effective root mean square transmit field, $B_{1,\text{rms}}^+$, and a predefined B_1^+ target, under constraints of peak and average RF power and SAR (local and global). Two configurations were evaluated: (i) protected mode (PM), in which conservative RF power limits are used to indirectly constrain SAR, and (ii) VOP mode, in which RF energy limits and explicit SAR limits are applied. In the VOP mode, the average RF power per channel was limited to 3 W, compared to 1.5 W in PM. The total power limit (sum across channels) in the VOP mode was set to 18 W. In the PM, this limit was 8 W. In the VOP mode only, local SAR constraints were set to 20 W/kg and global SAR was limited to 3.2 W/kg, corresponding to the “Level 1” SAR mode of IEC [33]. All optimizations were performed in MATLAB (MathWorks, version 24.1, R2024a).

For the imaging (excitation) module of the CEST sequence, dynamic RF shimming with three-spoke excitation [20] was employed for each slice.

The optimized saturation and excitation pulses were subsequently integrated into a two-dimensional multi-slice EPI CEST pulse sequence [18], for which the CEST saturation module was played once per repetition time (TR) right before the EPI slice selection.

For comparison purposes, pTx pulses were also optimized for 7 T simulations using the same power limits. The local SAR constraint was, however, set to 10 W/kg while the global SAR constraint remained 3.2 W/kg (“Normal” mode of IEC [33]).

2.2.2 | CEST Acquisition Parameters

For both in vitro and in vivo acquisitions, 5 slices (1.6 mm isotropic resolution, 0.3 mm slice gap) were acquired. The imaged field of view was $210 \times 210 \text{ mm}^2$, with a matrix size of 128×128 and GRAPPA [34] acceleration factor 2. The EPI bandwidth was set to 1698 Hz/Px. Two sets of CEST pulses (Gaussians of duration 50 ms) were designed to provide efficient saturation of the target protons while maintaining frequency selectivity: “low- B_1^+ ” pulses for amide, guanidinium, and hydroxyl protons, and “high- B_1^+ ” pulses for glutamate protons.

For the low- B_1^+ acquisitions, 12 saturation pulses (comprising 6 A-B pairs) were applied, resulting in an effective saturation time of 600 ms at a target $B_{1,\text{rms}}^+$ of 1.5 μT . Other imaging parameters were: TR = 2500 ms, echo time (TE) = 16 ms, flip angle (FA) = 73° . The acquisition duration was 2:53 min.

For the high- B_1^+ acquisitions, 6 saturation pulses (comprising 3 A-B pairs) were applied, resulting in an effective saturation time of 300 ms at a target $B_{1,\text{rms}}^+$ of 2.5 μT . Only for Participant 3 (P3), the $B_{1,\text{rms}}^+$ target was 3.0 μT and due to exceeded RF coil power

limits, the acquisition volume for this contrast was reduced to 4 slices. Due to the higher energy deposition of the high- B_1^+ CEST saturation module, the TR was increased to 5000 ms, resulting in an acquisition duration of 4:05 min. Other imaging parameters were: TE = 16 ms, FA = 85° .

To correct for static field inhomogeneities, Water Saturation Shift Referencing (WASSR) [35] scans were acquired throughout the protocol using either a high- B_1^+ or a low- B_1^+ pulse design, but with the pulse amplitude scaled down to $B_{1,\text{rms}}^+ = 0.3 \mu\text{T}$. Other imaging parameters were: TR = 1500 ms, TE = 16 ms, FA = 62° . The acquisition duration was 45 s.

To test the performance of the CEST saturation module, a 1.5 μT low- B_1^+ pulse design was evaluated on the PBS-based phantom, with the pulse amplitude reduced to achieve five $B_{1,\text{rms}}^+$ targets: 1.5, 1.4, 1.2, 1.0, and 0.5 μT . The imaging parameters were the same as for the low- B_1^+ acquisitions.

In vivo, the relayed nuclear Overhauser effect (rNOE), arising from aliphatic protons of mobile proteins and macromolecules, was imaged with the low- B_1^+ pulse amplitude reduced by 33% to obtain $B_{1,\text{rms}}^+ = 1.0 \mu\text{T}$. All other imaging parameters matched those for low- B_1^+ acquisitions.

For two participants (P2 and P4), an additional low- B_1^+ acquisition ($B_{1,\text{rms}}^+ = 1.5 \mu\text{T}$) with 16 saturation pulses was added to study the effect of prolonged saturation ($T_{\text{sat}} = 800 \text{ ms}$) on the CEST contrast. Other imaging parameters were: TR = 4500 ms, TE = 16 ms, FA = 73° . The repetition time had to be increased when increasing the saturation time to comply with SAR and system limits. The acquisition duration was 5:11 min.

All CEST acquisition parameters are listed in Table 1. For each in vivo CEST condition, with the exception of WASSR scans, two datasets were acquired and later averaged.

To enable distortion correction using Topup [36, 37], additional images were acquired during the in vivo sessions with the same imaging parameters as the low- B_1^+ acquisition, but without the CEST saturation module. One image was acquired with the same phase-encoding direction as the CEST acquisitions (anterior–posterior) and a second image was acquired with the reversed phase encoding (posterior–anterior). A T_1 -weighted anatomical MP2RAGE [38] image with 0.6 mm isotropic resolution was also acquired on each volunteer for brain segmentation purposes, with the following parameters: TR = 5100 ms, inversion times = 991 ms/2970 ms, bandwidth = 300 Hz/Px, FA = $4^\circ/3^\circ$, matrix size = 326×360 . The complete protocol lasted 1 h and 30 min.

2.3 | Data Processing

In vivo CEST acquisitions were motion corrected using FSL flirt [39]. The first volume of each acquisition was aligned across acquisitions, after which all volumes within each session were realigned to the corresponding session's first volume. Both stages employed rigid-body transformation. From the two scans acquired with opposite phase encoding directions, a

TABLE 1 | CEST acquisition parameters.

	Low- B_1^+		High- B_1^+	WASSR
	12 pulses	16 pulses		
TR (ms)	2500	4500	5000	1500
TE (ms)		16	16	16
$B_{1,rms}^+$ (μ T)	1.0; 1.5	1.5	2.5 (P2, P4); 3.0 (P3)	0.3
T_{sat} (ms)	600	800	300	300; 600
$\Delta\omega$ (ppm)	-40, -40, -40, -10, -5.0:0.2:0, 40, 40, 40, 10, 5.0:-0.2:0	-40, -40, -40, -10, -5.0, -4.5, -4.0, -3.8, -3.5, -3.2:0.2:-2.4, -2.0, -1.8, -1.5, -1.0:0.2:0, 40, 40, 40, 10, 5.0, 4.5, 4.0, 3.8, 3.5, 3.2:-0.2:2.4, 2.0, 1.8, 1.5, 1.0:-0.2:0		1.0, 1.0, 1.0:-0.1:0, -1.0, -1.0, -1.0:0.1:0
T_{acq} (min:sec)	2:53	5:11	4:05	0:45

Abbreviations: $\Delta\omega$, saturation offsets; T_{acq} , acquisition time; TE, echo time; TR, repetition time; T_{sat} , saturation time.

susceptibility-induced off-resonance field was estimated using FSL Topup [36, 37] and was applied to each CEST volume to correct for EPI distortions ranging up to ± 2 voxels over our cohort.

For both in vitro and in vivo acquisitions, voxel-wise frequency offset maps obtained from WASSR acquisitions were used to correct B_0 inhomogeneities by interpolating each CEST Z-spectrum to the corrected frequency offsets. Repeated in vivo CEST acquisitions were then averaged to improve the signal-to-noise ratio and cubic-interpolated to 0.1 ppm spacing between -5 and 5 ppm.

To generate the different CEST contrast maps, the magnetization transfer ratio asymmetry [40] (MTR_{asym}) at a given frequency offset $\Delta\omega$ was computed on a voxel-by-voxel basis:

$$MTR_{asym}(\Delta\omega) = \frac{S(-\Delta\omega) - S(\Delta\omega)}{S_{ref}} \quad (1)$$

where S is the acquired CEST signal and $S_{ref} = S(-40 \text{ ppm})$ is the unsaturated reference signal used for normalization.

For both in vitro and in vivo acquisitions, hydroxyl- and glutamate-weighted contrasts were generated by computing MTR_{asym} at five frequency offsets (0.8:0.1:1.2 ppm for hydroxyl protons and 2.8:0.1:3.2 ppm for glutamate protons) and averaging the resulting MTR_{asym} maps for each contrast. For each volunteer, APT-weighted, guanidinium-weighted, and rNOE-enhanced maps were generated by computing MTR_{asym} at discrete frequency offsets within defined ranges, 3.5 ± 0.1 , 2.2 ± 0.2 , and -3.5 ± 0.5 ppm, respectively, and averaging the resulting maps.

Additionally, for both APT-weighted and guanidinium-weighted contrasts, Lorentzian difference analysis (LDA) [41] was performed to remove magnetization transfer (MT) and direct saturation (DS) effects to better reflect the contribution of each pool. Specifically, DS and MT were fitted using Lorentzian functions, and the resulting spectrum (reference spectrum) was subtracted from the normalized Z-spectrum. The residual signal was then integrated over the same frequency offsets as those

used for MTR_{asym} . The initial conditions and boundaries of the fits are listed in Table S1.

Gray matter (GM) and white matter (WM) segmentations were performed on the T_1 -weighted images using sMRIprep [42] (version 0.18.0) and co-registered with the CEST images using custom Matlab scripts. For each subject, median MTR_{asym} and LDA values were computed within GM and WM ROIs. Group results are reported as the mean across subjects.

3 | Results

3.1 | Simulations

In vitro normalized circularly polarized (CP) B_1^+ profiles obtained at 7 and 11.7 T are shown in Figures 1A and 1B, respectively. To evaluate the effectiveness of the pTx pulse design in mitigating B_1^+ inhomogeneities at 11.7 T and to compare it with that at 7 T, the normalized root mean square error (nRMSE) was computed for both the protected and the VOP modes as a function of $B_{1,rms}^+ \sqrt{DC}$, where $DC = T_{sat}/TR$ is the duty cycle (Figure 1C). For further comparison, the performance of the CP design under the same energy management strategies is also shown in Figure 1D. The results obtained for 7 T are consistent with previously reported findings [18, 23]. Across all pulse design strategies, 7 T optimization consistently yielded lower nRMSE values than at 11.7 T. This difference can be attributed to the more homogeneous initial CP B_1^+ profile at 7 T (Figure 1A) compared to that at 11.7 T (Figure 1B), which facilitates convergence toward lower nRMSE values during optimization. At 11.7 T, the use of VOP-based SAR constraints was essential to mitigate transmit field inhomogeneities, resulting in an average reduction of the pTx pulses nRMSE by 13% compared to the PM constraints across the total $B_{1,rms}^+ \sqrt{DC}$ range tested. For $B_{1,rms}^+ \sqrt{DC}$ values up to 1 μ T, 11.7 T pTx pulses designed under VOP-based SAR constraints achieved similar performance to 7 T pulses, yielding average nRMSE values of 6.3% (vs. 4.6% at 7 T). Given the superior performances of VOP-SAR-constrained pulses, in vitro and in vivo acquisitions were only conducted with pulses optimized under these constraints.

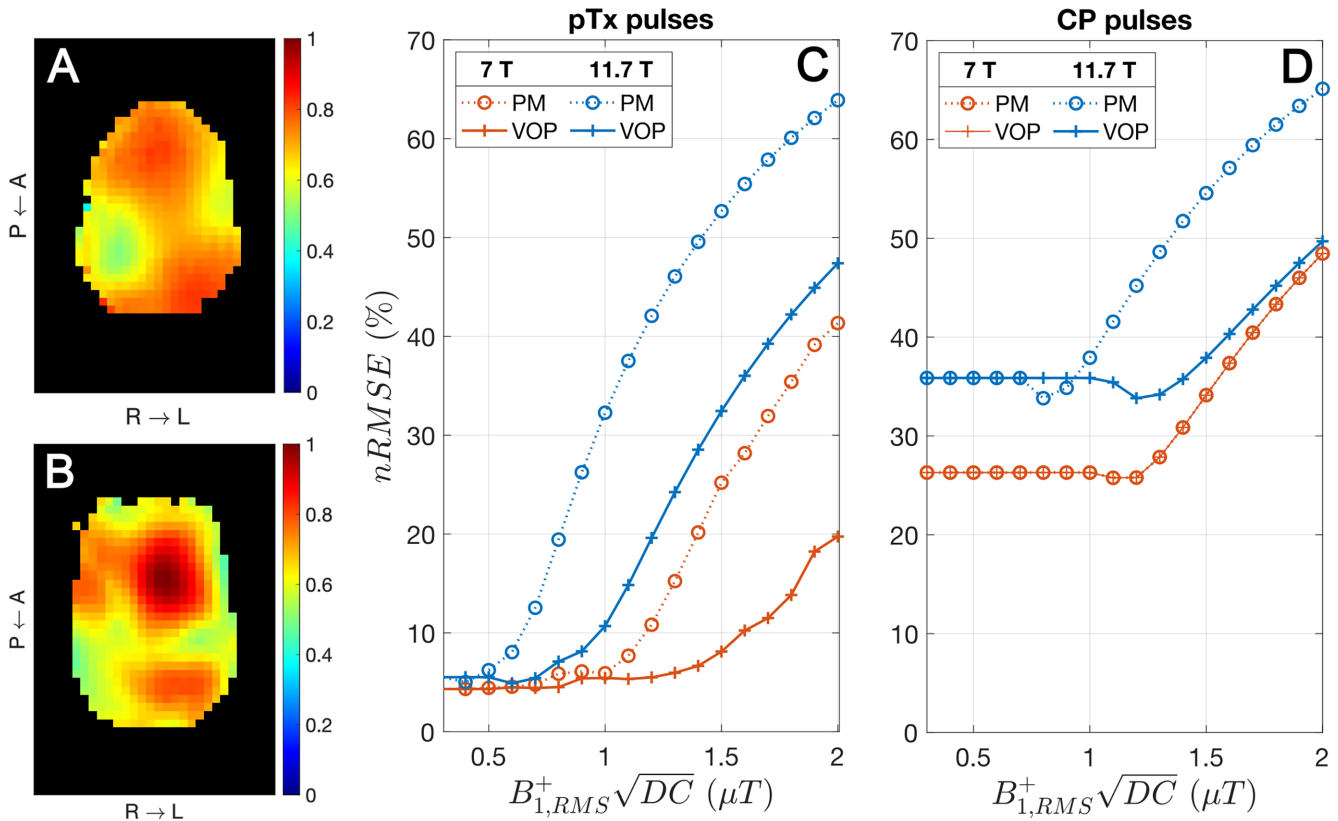


FIGURE 1 | Normalized circularly polarized B_1^+ profiles at 7 T (A) and 11.7 T (B) on a gelatin-based human-head-shaped phantom. Simulations of the normalized root mean square error (nRMSE) for tailored parallel transmission (pTx) (C) and for circularly polarized (D) pulses at 7 and 11.7 T, optimized under protected mode (PM) or virtual observation points (VOP) mode, are also shown.

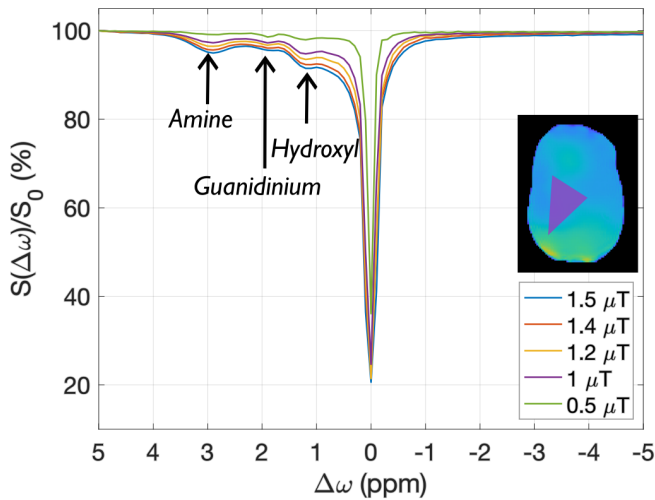


FIGURE 2 | Z-spectra averaged over a region of interest (purple triangle), acquired on the PBS-based phantom at different $B_{1+,rms}$ amplitudes, showing clearly identifiable resonance peaks corresponding to glucose-hydroxyl, creatine-guanidinium, and glutamate-amine protons.

3.2 | In Vitro

Experiments performed on the PBS-based phantom demonstrated clear spectral separation of glutamate, guanidinium, and hydroxyl protons across five B_1^+ amplitudes (Figure 2).

TABLE 2 | In vivo CEST pulse design performances.

	P1 nRMSE (%)	P2 nRMSE (%)	P3 nRMSE (%)	P4 nRMSE (%)
Low- B_1^+ (1.5 μT)	4.5	12.1	8.4	4.3
High- B_1^+ (2.5 μT)		6.3		5.3
High- B_1^+ (3.0 μT)			12	

Abbreviations: P, participant; nRMSE, normalized root mean square error.

Saturation homogeneity was evaluated on the gelatin-based phantom and showed moderate spatial variations, attributable to experimental factors inherent to ultra-high field phantom imaging (additional details are given in the Supporting Information and Figure S1).

3.3 | In Vivo

For the in vivo experiments, the optimized pTx-VOP mode CEST pulses yielded nRMSE values ranging from 4.3% (P4, low- B_1^+ pulses) to 12.1% (P2, low- B_1^+ pulses) (Table 2). Mean Z-spectra over WM and GM ROIs for the two saturation amplitudes are presented for all participants in Figure 3. MTR_{asym} maps demonstrated a clear contrast between GM and WM across all participants, in agreement with the T_1 -weighted images. MTR_{asym} maps of multiple CEST contrasts across all participants are shown in

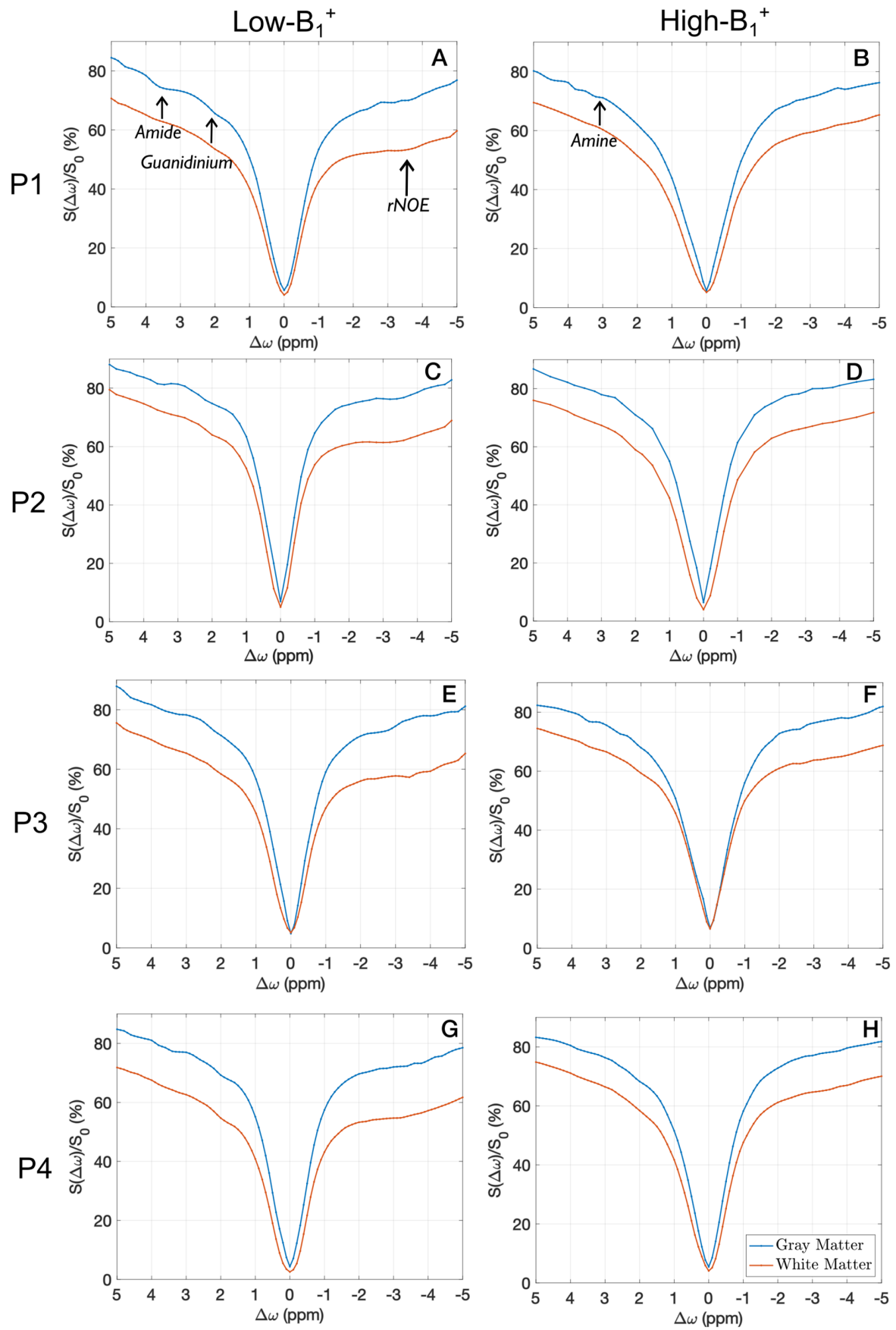


FIGURE 3 | Z-spectra averaged on gray matter and white matter regions of interest for each participant for low- B_1^+ acquisitions (A, C, E, G) and high- B_1^+ acquisitions (B, D, F, H).

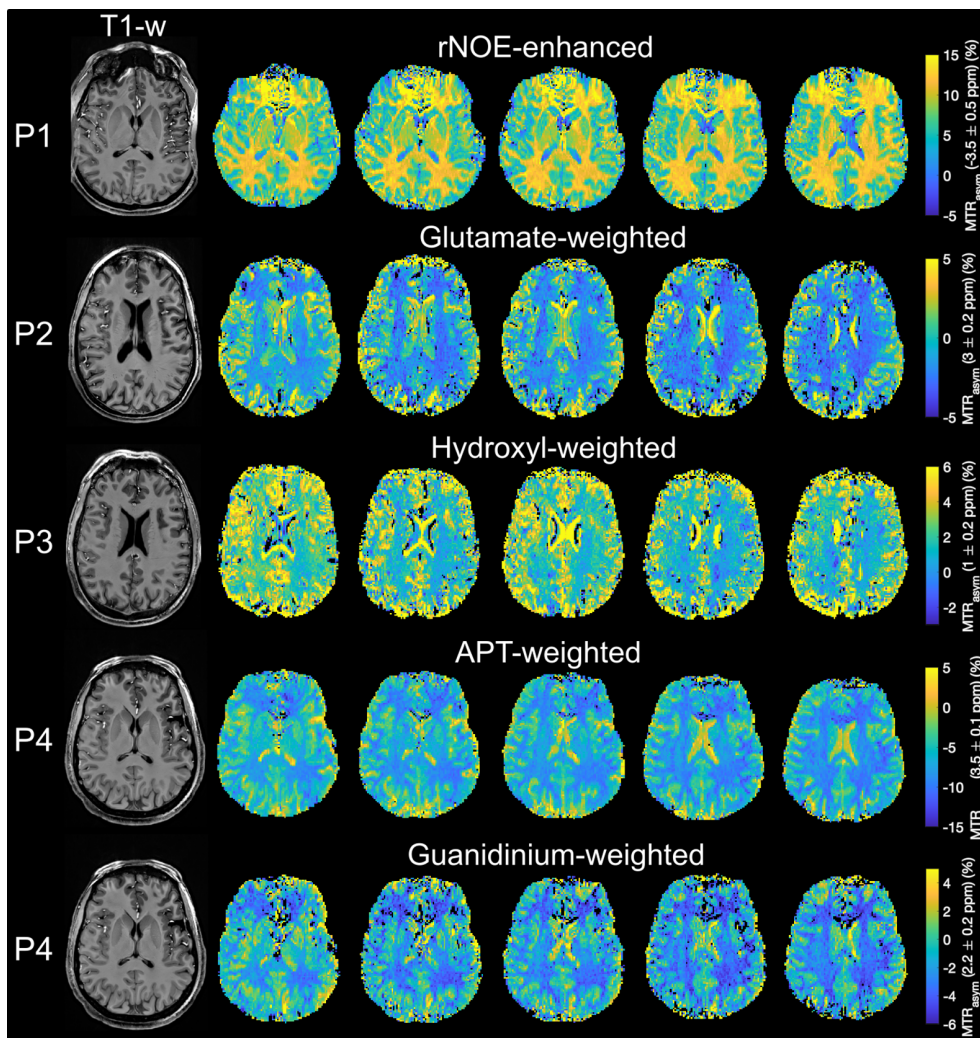


FIGURE 4 | CEST MTR_{asym} maps for four different participants and different contrasts. Each row represents a different CEST contrast. From top to bottom: rNOE-enhanced CEST (Participant 1), glutamate-weighted CEST (Participant 2), hydroxyl-weighted CEST (Participant 3), APT-weighted CEST (Participant 4), and guanidinium-weighted CEST (Participant 4). In the first column are shown the T_1 -weighted anatomical images corresponding to the first slice of the acquired CEST volume.

Figure 4 and all acquired CEST contrasts for P3 are shown in Figure S2. CEST GM–WM contrasts were consistent across all participants, as reflected by the voxel-wise MTR_{asym} distributions in Figure 5. Across all subjects, the GM contrast for hydroxyl-, glutamate-, guanidinium- and APT-weighted CEST ranged from 1.5% to 2.6%, -0.2% to 1.5% , -1.7% to -0.6% , -5.3% to -4.6% , respectively. WM contrast for rNOE-enhanced maps ranged from 10.1% to 11.4% for all participants.

Among the evaluated contrasts, rNOE-enhanced maps exhibited the strongest tissue differentiation, with a mean $\Delta MTR_{\text{asym}}(\text{WM} - \text{GM})$ of 4.3% across all participants (range: 3.5% – 5.0%). All acquired rNOE-enhanced MTR_{asym} maps are shown in Figure S3. Hydroxyl-weighted CEST demonstrated the weakest tissue contrast, with a mean $\Delta MTR_{\text{asym}}(\text{GM} - \text{WM})$ of 0.7% (range: 0.5% – 1.1%). APT-weighted maps showed an average $\Delta MTR_{\text{asym}}(\text{GM} - \text{WM})$ of 4.1% (range: 3.4% – 4.4%) while guanidinium-weighted CEST yielded a more moderate contrast ($\Delta MTR_{\text{asym}}(\text{GM} - \text{WM}) = 2.3\%$) with rather small variability over the cohort (range: 2.2% – 2.5%). We note that using the MTR_{asym}

metric, both the APT and guanidinium contrasts are influenced by rNOE, which has an important contribution between -2 and -4 ppm for low- B_1^+ saturation acquisitions (Figure 3A). The LDA analysis mitigates these competing effects and thus provides a more specific estimate of the CEST contrast within this spectral range. Using this approach, the average $\Delta \text{LDA}(\text{GM} - \text{WM})$ yields values of 0.3% (range: 0.1% – 0.4%) for both APT- and guanidinium-weighted CEST on our cohort. LDA maps for P4 are shown in Figure 6. Median MTR_{asym} and LDA values in GM and WM for both contrasts, for all participants are listed in Table S2.

Increasing the saturation duration to 800 ms enhanced both guanidinium- and hydroxyl-weighted $\Delta MTR_{\text{asym}}(\text{GM} - \text{WM})$ contrast by $+0.3\%$ in P2, accompanied by an increase of $MTR_{\text{asym}}(\text{GM})$ of $+1.4\%$ for both contrasts. The guanidinium-weighted MTR_{asym} maps for $T_{\text{sat}} = 600$ ms and $T_{\text{sat}} = 800$ ms are shown in Figure 7A,B, respectively. The voxel-wise distributions of $MTR_{\text{asym}}(\text{GM})$ values for guanidinium-weighted CEST in P2 at both saturation times are shown in Figure 7C, demonstrating

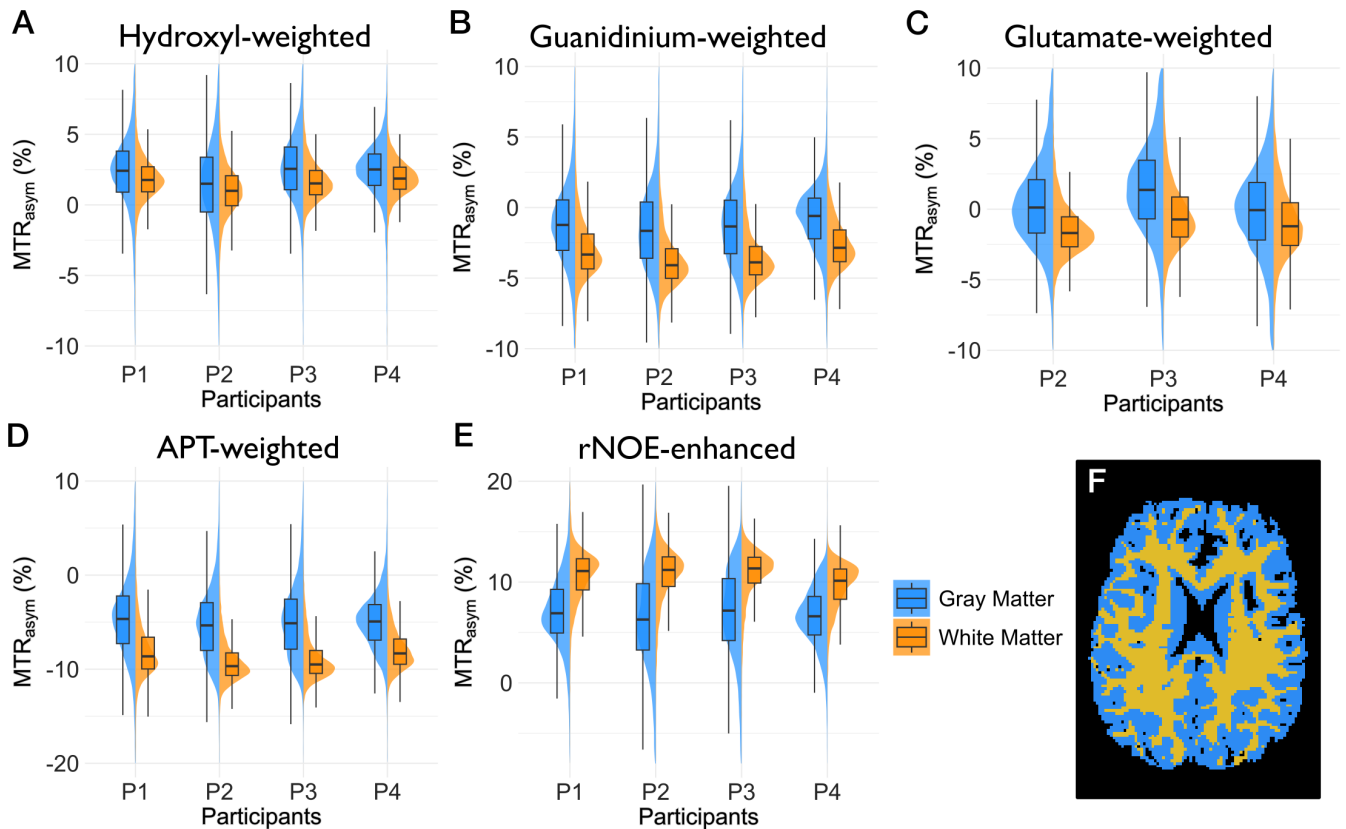


FIGURE 5 | Voxel-wise distributions of gray matter (blue) and white matter (orange) MTR_{asym} values for the four participants for hydroxyl-weighted (A), guanidinium-weighted (B), glutamate-weighted (C), APT-weighted (D) and rNOE-enhanced (E) CEST contrasts. An example of gray matter (blue) and white matter (orange) segmentation on Participant 4 is shown in (F).

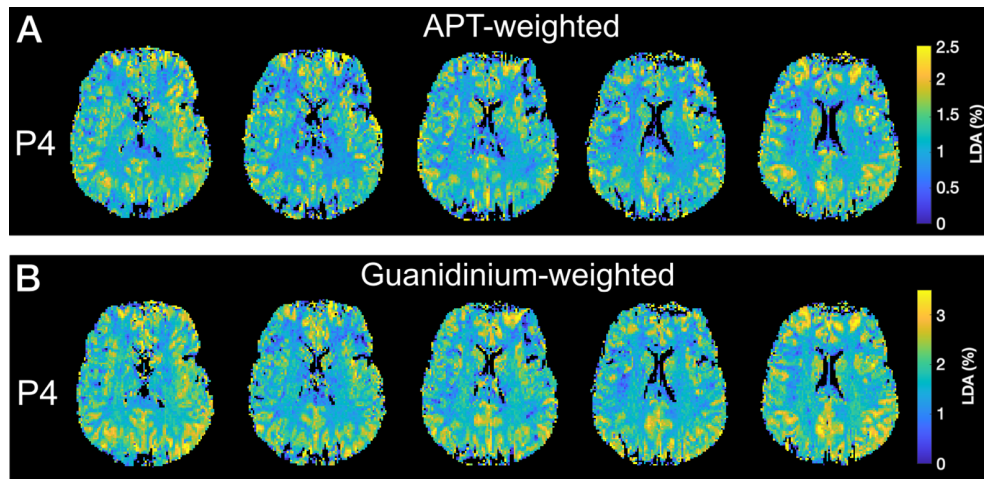


FIGURE 6 | Lorentzian difference analysis (LDA) for APT-weighted CEST (A) and guanidinium-weighted CEST (B) on Participant 4.

a shift toward higher MTR_{asym} values with increased saturation time, while the standard deviation remained unchanged. For P4, extending the saturation time resulted in only a modest gain in hydroxyl-weighted $\Delta MTR_{asym}(GM-WM)$ contrast of +0.2% and no measurable improvement for guanidinium-weighted CEST contrast, but for both cases, the median $MTR_{asym}(GM)$ increased by approximately 0.9%. Longer saturation times led to a slight decrease in APT-weighted $\Delta MTR_{asym}(GM-WM)$ contrast for both participants (P2: -0.7% and P4: -0.1%), primarily driven by

moderate increases in the median $MTR_{asym}(GM)$ values (+0.2% for P2 and +0.5% for P4).

Finally, glutamate-weighted CEST acquired at a saturation amplitude of 2.5 μT (Figure 4) yielded contrast values ($\Delta MTR_{asym}(GM-WM)$) of 1.8% for P2 and 1.0% for P4. Increasing the B_1^+ amplitude further enhanced the contrast, reaching a $\Delta MTR_{asym}(GM-WM)$ of 2.2% in P3 but also increasing the nRMSE to 12% (Table 2).

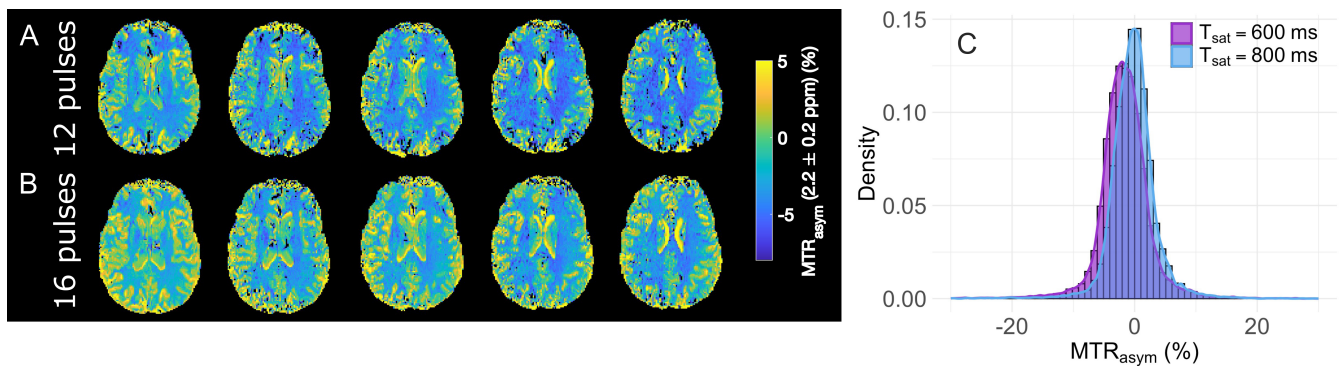


FIGURE 7 | Effect of extended saturation time on guanidinium-weighted CEST MTR_{asym} maps in Participant 2. CEST maps corresponding to T_{sat} of 600 ms (A) and 800 ms (B) are shown. For this contrast, the distributions of the MTR_{asym} in gray matter voxels indicate an increased CEST effect with longer T_{sat} (C).

A Z-spectrum-based two-point B_1^+ post-processing correction [28] was evaluated for the 1.5 μT datasets of each participant using acquisitions at 1.0 and 1.5 μT . Overall, this correction led to only marginal improvements in CEST contrast across participants and contrast weightings. The median MTR_{asym} values in GM and WM pre-correction and post-correction are listed in Table S3 for all participants, for all evaluated contrasts. The largest post-correction differences were observed for P2, who also exhibited the highest pre-correction nRMSE (12.1%). Localized improvements in GM–WM contrast were especially seen in the left supramarginal and left superior parietal regions (Figure 8A,B, highlighted in red). The corresponding difference maps between images in Figure 8A,B are shown in Figure 8C. Across the entire imaged volume for this participant, the guanidinium-weighted MTR_{asym} increased by 0.8% in WM and 0.5% in GM; however, the overall ΔMTR_{asym} (GM–WM) contrast decreased by 0.4%. For all the other participants, the differences were negligible. As an example, Figure 8D shows the difference between B_1^+ -corrected and uncorrected guanidinium-weighted MTR_{asym} maps in P1, demonstrating a highly homogeneous difference map with values close to zero.

4 | Discussion

This study reports the first in vivo human brain CEST images acquired at 11.7 T, demonstrating the feasibility of the method at this field strength. A tailored interleaved pTx CEST pulse design under VOP-based SAR constraints was employed to mitigate transmit field inhomogeneities. Following initial testing on metabolite-doped PBS and gelatin phantoms, multiple contrast weightings were acquired at different saturation amplitudes in four healthy young volunteers, yielding measurable and consistent CEST effects across the cohort despite the increased technical challenges associated with ultra-high field MRI.

One of the main motivations for performing CEST MRI at high magnetic fields is the increased selectivity and detectability of metabolites [17], driven primarily by the larger chemical shift dispersion between exchanging proton pools (in Hz). This increased dispersion reduces spectral overlap and spillover effects, enabling a finer inspection of metabolite-related features in the Z-spectrum. In the present study, the low- B_1^+ Z-spectra

show clearly resolved dips corresponding to amide and guanidinium resonances, which are typically barely discernible at 7 T [43]. No additional CEST resonances beyond those established at lower field strengths were identified over the investigated frequency range, though we note that the acquisition parameters were optimized for robust imaging of known metabolites rather than systematic spectral exploration.

The acquired CEST images exhibited clear gray-white matter differentiation across multiple CEST contrasts corresponding to different saturation offsets. The highest GM–WM contrasts were observed in MTR_{asym} -based APT-weighted and rNOE-enhanced images, with values comparable to previous studies at 7 T [12, 18, 41, 44]. Higher rNOE-enhanced contrast values have been reported using a relaxation-compensated metric [5]. MTR_{asym} -based guanidinium-weighted CEST imaging at 11.7 T revealed clear tissue differentiation across all participants, with GM–WM contrast exceeding that reported in previous studies [25, 43, 45]. Following LDA, both APT and guanidinium GM–WM contrasts were reduced compared to MTR_{asym} analysis, as expected given the rNOE contamination inherent to asymmetry-based analyses. Nonetheless, GM signal remained higher than WM signal across all subjects for both pools (APT: GM 1.3%–1.7% vs. WM 1.0%–1.5%; Guanidinium: GM 2.0%–2.3% vs. WM 1.6%–2.1%), consistent with known higher concentrations of cellular proteins and creatine in GM [46–50]. Glutamate-weighted GM–WM contrast was moderate but remained satisfactory, particularly at $B_{1,\text{rms}}^+ = 3 \mu\text{T}$, with values close to those recently reported at 7 T [51] with longer saturation times (800 ms). Human brain imaging of resonant metabolites around 1 ppm has been reported in only a few studies, and while hydroxyl-weighted images showed the weakest contrast at 11.7 T, the observed values still surpassed those previously reported [18, 52].

Absolute CEST signal values were sometimes lower than those reported in prior studies, likely due to the shorter saturation times used in the present work. For example, APT-weighted CEST studies typically employ saturation durations between 1 and 3 s, and report correspondingly larger signal amplitude values [12, 53, 54]. Guanidinium-weighted CEST at 2 ppm has also been reported using a saturation time of 1500 ms, yielding higher CEST signal intensity values; however, these measurements were restricted

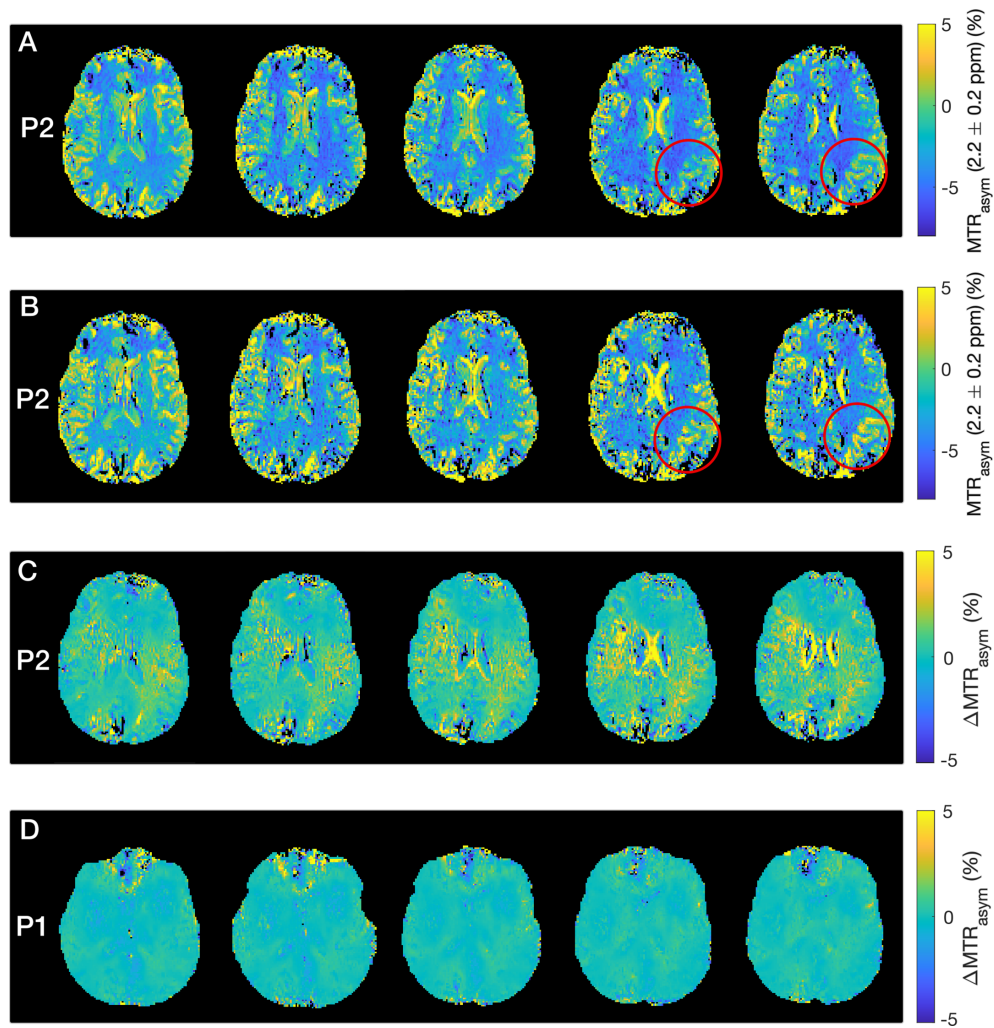


FIGURE 8 | The post-processing B_1^+ correction did not yield measurable improvements in any participant other than Participant 2. For this participant, the corrected guanidinium-weighted CEST MTR_{asy} maps (B) show localized gray–white matter contrast enhancement (red circles) compared with the uncorrected maps (A). The MTR_{asy} difference maps for Participant 2 (C) exhibit larger correction-induced changes, whereas Participant 1, as an example, shows minimal differences (D).

to localized regions of interest and did not assess whole-brain GM–WM contrast [44]. In the present study, increasing the saturation time from 600 to 800 ms yielded modest improvements with higher GM CEST signal intensity values in APT-, hydroxyl- and guanidinium-weighted CEST but no significant changes on the GM–WM contrast. For high- B_1^+ acquisitions, as previously suggested in the literature [16], increasing the B_1^+ target to 3 μ T improved CEST signals and GM–WM contrast but this was achievable only with a reduced slice coverage due to the coil protection limits being exceeded and also resulted in increased saturation inhomogeneity. These constraints on saturation duration and B_1^+ amplitude remain one of the main challenges for CEST imaging of the human brain at ultra-high field.

Despite these limitations, the present acquisition conditions allowed clear detection of anatomical contrast at 11.7 T, highlighting the potential of ultra-high field CEST for targeted studies of localized brain regions in neurological diseases. At the same time, quantitative interpretation across studies remains complicated by the strong dependence of the CEST contrast on both tissue properties (pH, temperature, conductivity...)

and acquisition-specific factors, including saturation scheme and field strength, which vary substantially across the literature [55–57].

In this work, the MTR_{asy} metric was selected as a simple and low-assumption approach for CEST contrast quantification. As previously discussed, this metric reflects overlapping contributions from multiple effects, including rNOE and magnetization transfer from macromolecules such as proteins and peptides. These confounding contributions are particularly important for guanidinium and APT contrasts; LDA [41] analysis was therefore performed for these two contrasts to improve specificity. More advanced quantification metrics based on voxel-wise fittings such as multi-pool Lorentzian fitting [58], polynomial and Lorentzian line-shape fitting [59] have been proposed. While such methods may enable improved separation of individual CEST contributions, they are highly sensitive to the choice of initial fitting values and boundary conditions, which can lead to substantial variability in the resulting contrast and require careful interpretation [60]. Moreover, for high- B_1^+ power, Lorentzian fitting approaches are not well-suited as the assumption of no interaction between

the pools breaks down [55]. The systematic evaluation of such additional post-processing strategies was beyond the scope of the present study, which was primarily designed to establish the technical feasibility of CEST MRI on an 11.7 T system for humans and to acquire robust and reliable CEST images of the human brain. Nevertheless, quantitative CEST represents a particularly compelling avenue for future work at this field strength, where the increased chemical shift dispersion and sensitivity could be leveraged for more accurate estimation of metabolite concentrations and exchange rates [61, 62]. T_1 -normalized metrics such as AREX [5] could additionally improve specificity by removing T_1 contributions to the CEST contrast. In our healthy cohort, AREX-derived CEST maps showed spatial patterns highly consistent with MTRasym-derived maps on all participants, for all contrasts (see Figure S4 showing AREX maps for Participant 3). However, AREX may be particularly relevant in pathological conditions where T_1 could vary substantially across tissue types.

As expected, the B_1^+ inhomogeneities were more pronounced at 11.7 T than at lower field strengths. Nonetheless, by allowing a higher local SAR (20 vs. 10 W/kg) the pTx-VOP mode enabled transmit field homogenization comparable to previously reported 7 T CEST optimizations [18, 23], resulting in acceptable saturation homogeneity (typically smaller than 12.1% over a 1 cm slab). The retrospective B_1^+ correction provided minimal gains overall with the exception of very localized improvements observed in a single participant, consistent with previous observations by Völke et al. [25]. These results suggest that, in the context of effective pTx optimization, retrospective B_1^+ correction may offer limited additional benefits. This finding is particularly important for future high-field studies in patient cohorts, for which avoiding the need for additional datasets is critical to minimize total scan time.

In this study, saturation pulses were tailored on a per-subject basis to account for inter-individual B_1^+ variability; the development of universal pTx pulse designs applicable across subjects would substantially simplify clinical research deployment. Future improvements in pTx RF coil design may help achieve a more homogeneous B_1^+ field, which in turn may decrease the SAR and energy demand of CEST saturation and excitation pulses and allow more flexibility for the CEST saturation parameters.

5 | Conclusion

This work demonstrates the feasibility of performing multi-contrast CEST MRI in the human brain at 11.7 T with 1.6 mm isotropic resolution using pTx-optimized CEST saturation and excitation pulses. The acquisition and optimization framework, based on pTx saturation pulse design under VOP-based SAR constraints and previously established at 7 T [18], was successfully translated to 11.7 T. The primary technical adaptations were the use of higher local SAR limits within the IEC Level 1 operating mode and a dedicated B_0 shimming protocol. By combining custom sequence implementation with tailored pTx saturation pulse design, we achieved robust and reproducible gray-white matter CEST contrast across multiple contrast weightings in healthy volunteers despite the stringent RF and SAR constraints. Importantly, at 11.7 T, the combination of higher field strength and our

optimized experimental conditions produced clear gray-white matter contrast, which may provide a more informative readout than single-compartment CEST signals alone. CEST acquisitions at 11.7 T remain limited by B_1^+ inhomogeneity and coil protection constraints; however, the pTx-VOP mode effectively mitigates these challenges, enabling practical in vivo imaging. Further advances in RF hardware, saturation pulse design and optimization will improve performance and spatial coverage, thereby fully realizing the potential of ultra-high field CEST MRI.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Human head-shaped phantom used for the experiments (A) and corresponding unsaturated CEST images (B). Simulated $B_{1,\text{rms}}^+$ maps for hydroxyl-weighted (C, $B_{1,\text{rms}}^+ = 1.5 \mu\text{T}$) and glutamate-weighted (E, $B_{1,\text{rms}}^+ = 2.5 \mu\text{T}$) CEST acquisitions on the gelatin-based phantom. The acquired CEST maps are shown in D (hydroxyl-weighted) and F (glutamate-weighted). The difference in slice number between the $B_{1,\text{rms}}^+$ and CEST maps results from the difference in slice thickness and slice gap (2.5 mm thickness with 2.5 mm gap vs. 1.6 mm thickness and 0.3 mm gap). **Figure S2:** CEST MTR_{asym} maps acquired on Participant 3: rNOE-enhanced (A), hydroxyl-weighted (B), APT-weighted (C), guanidinium-weighted (D), glutamate-weighted (E). The $B_{1,\text{rms}}^+$ target for the glutamate-weighted acquisition on this participant was 3 μT (compared to 2.5 μT for the other participants) and coil protection limits were reached; therefore, the slice volume was reduced to 4 slices instead of 5. **Figure S3:** rNOE-enhanced MTR_{asym} maps acquired on all participants. **Figure S4:** AREX-based CEST contrasts for Participant 3: rNOE-enhanced (A), hydroxyl-weighted (B), APT-weighted (C), guanidinium-weighted (D) and glutamate-weighted (E). **Table S1:** Fitting boundaries and initial values for the Lorentzian difference analysis. The unit of peak width (Γ) and peak offset (δ) is ppm. **Table S2:** Median $\pm$ standard deviation of MTR_{asym} and Lorentzian difference analysis (LDA) values in white matter and gray matter for all participants, for APT- and guanidinium-weighted contrasts. **Table S3:** Median $\pm$ standard deviation of MTR_{asym} values in white matter (WM) and gray matter (GM) before and after B_1^+ correction, for all participants, for all tested contrasts.